Vasu Mekala

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PROFESSIONAL SUMMARY

Applied AI Engineer with 4 years of experience turning data and models into products that teams use every day. I built the AI and backend components of a budget forecasting app and designed a multi-tenant PostgreSQL data layer with 81 tables and row-level security. I also built churn and propensity models on 500K+ records that guided retention targeting and cut data prep time by 35%. I track drift so models get flagged for retraining before they ship.

WORK EXPERIENCE

Executive Landscaping Inc. | AI Engineer | North Haven, CT Oct 2026 - Present

- Building AI-powered internal apps for planning, forecasting, and business automation, including a budget forecasting platform used by 300 people across forecasting cycles that take 2 to 3 weeks.
- Evaluating GenAI options in Azure AI Foundry, including GPT-5-mini and embedding models, and checking access, quotas, networking, and deployment needs before production.
- Own the AI application lifecycle from requirements and system design through backend, data modeling, testing, deployment, and monitoring, supporting 2 to 4 releases per month.
- Design and maintain a multi-tenant PostgreSQL data layer with 81 tables and 75 row-level security policies, so each client sees only its own data.
- Write TypeScript financial logic and backend APIs, backed by 240+ automated tests and Playwright end-to-end tests running in CI.
- Handled nearly 16 production outages using telemetry analysis and health and database alerts, and scoped upgrades for a CRM assistant with speech-to-text and Microsoft Teams integration.

Verizon | Data Scientist | Contract (Remote) May 2025 - Aug 2026

- Built churn and propensity models in Python with Scikit-learn and XGBoost, improving retention targeting by 5 to 10 percent.
- Processed 500K+ records with SQL, PySpark, and Databricks, cutting feature prep effort by 35%.
- Compared models on ROC-AUC, precision-recall, calibration, and cross-validation. Used gradient boosting for the nonlinear patterns.
- Analyzed NPS and campaign results with regression, hypothesis testing, and A/B testing, then shared segments with marketing.
- Logged experiments in MLflow and registered approved models in the Model Registry for handoff.
- Watched feature and prediction drift against training data and tested model changes in CI/CD before release.

University of New Haven | ML Research Assistant | West Haven, CT Nov 2023 - May 2024

- Supported ML research with data prep, exploratory analysis, feature engineering, and baseline experiments in Python.
- Read academic papers to choose methods and baselines, then tested them with validation metrics and error analysis.
- Contributed data preparation, model evaluation, and technical documentation used in faculty-led research and final project presentations.

HCLTech | Data Analyst, Healthcare Analytics | India Feb 2022 - Aug 2023

- Prepared model-ready datasets from 2M+ claims, encounters, admissions, eligibility, discharge, and pharmacy records using SQL, PySpark, Pandas, and NumPy.
- Helped build readmission-risk models with Scikit-learn using prior admissions, chronic conditions, ER use, medication activity, and care gaps. Used SHAP to explain what drove each prediction.
- Built ARIMA and Prophet forecasts for seasonal healthcare utilization to support capacity planning discussions.
- Automated recurring claims, billing, and data quality checks in Python and SQL, saving more than 20 hours of manual checking each week.

- Checked model behavior across patient risk and demographic groups, catching inconsistent predictions before results reached other teams.
- Worked with healthcare, data engineering, and QA teams on validation and HIPAA-aligned analytics before release.

SKILLS

Programming and Data: Python, SQL, TypeScript, PostgreSQL, PySpark, Pandas, NumPy

Machine Learning: Scikit-learn, XGBoost, Gradient Boosting, PyTorch, BERT, Hugging Face Transformers, Feature Engineering, Time-Series Forecasting

Generative AI: Large Language Models, Retrieval-Augmented Generation, LangChain, LlamaIndex, Prompt Engineering

MLOps and Cloud: MLflow, Model Deployment, Drift Detection, Databricks, Azure AI Foundry, FastAPI, Playwright, CI and CD

Statistics and Evaluation: A and B Testing, Hypothesis Testing, ROC-AUC, Cross-Validation, SHAP

PROJECTS

Enterprise Generative AI Knowledge Assistant

- Built a RAG knowledge assistant with LangChain and LlamaIndex that answers questions from policy and technical documents and shows the source passages behind each answer. I wrote the Python pipelines for chunking and semantic retrieval, served responses through FastAPI, and used MLflow to track prompt versions and retrieval quality.

Statistical Analysis of Claims and Communications Patterns

- Used BERT and statistical analysis to sort healthcare claims and member messages into clinical, billing, and eligibility groups. I pulled patterns from unstructured text with named entity recognition, then used precision-recall and significance testing to find where the classifier was weak and what to fix.

EDUCATION

University of New Haven Aug 2023 - May 2025
Master of Science, *AI and ML* GPA: 3.7

CERTIFICATIONS & ACHIEVEMENTS

- Azure AI Engineer Associate (AI-102) - Microsoft Sep 2025
- IBM Generative AI Engineering Professional Certificate - IBM May 2025
- Deep Learning Specialization - DeepLearning.AI May 2022